

International health data trends in anxiety-related and depression-related visits to primary care before, during, and after the COVID-19 pandemic: a cross-sectional study by INTRePID



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Summary

Background The COVID-19 pandemic forced primary care to adapt, balancing routine care with rising mental health demands. We examined international trends in anxiety-related and depression-related visits to primary care to better understand population needs during and after the pandemic era.

Methods This retrospective cross-sectional study by the International Consortium of Primary Care Big Data Researchers (INTRePID) examined anxiety-related and depression-related visits to primary care among adults aged 20 years and older in Australia, Brazil, Canada, China, Indonesia, Mexico, Norway, Peru, Singapore, Sweden, and the USA, from Jan 1, 2018 to Dec 31, 2023. No visits were excluded based on patients' sociodemographic characteristics or comorbidities, and all available visits were analysed. Aggregated data sourced from electronic medical records and administrative databases were stratified by sex and psychosocial life stages (20–29 years, 30–49 years, 50–69 years, and 70 years and older). We calculated the proportion of all primary care visits that were attributed to anxiety and depression. Generalised linear models estimated changes in visit rates across three periods: pre-pandemic (Jan 1, 2018–Mar 31, 2020), pandemic (April 1, 2020–Dec 31, 2021), and recovery (Jan 1, 2022–Dec 31, 2023). In China, periods were defined as: pre-pandemic (Jan 1, 2018–Dec 31, 2019), pandemic (Jan 1, 2020–Sept 30, 2021), and recovery (Oct 1, 2021–Dec 31, 2023).

Findings We examined 1 059 101 897 primary care visits from 11 countries, of which 49 022 341 (4.6%) visits were attributed to anxiety and depression among adults aged 20 years and older. The highest proportions of anxiety-related and depression-related primary care visits were observed in Canada, with 18 240 011 (8.1%) of 224 512 695 visits, and in the USA, with 451 067 (11.3%) of 3 998 109 visits. Visit rates increased during the pandemic compared with the pre-pandemic period, with the largest rise observed in Peru (RR 2.17 [95% CI 1.96–2.40]) and the smallest, non-significant increase in Sweden (RR 1.03 [0.96–1.10]). During the recovery period, visit rates in Canada, China, Peru, Sweden, and the USA remained significantly elevated compared with pre-pandemic levels (ranging from RR 1.91 [95% CI 1.65–2.22] in Peru to RR 1.18 [1.07–1.30] in Sweden). The highest visit rates were observed in females, ranging from a significant increase in Mexico (rate ratio [RR] 1.88 [95% CI 1.69–2.10]) to a non-significant change in Indonesia (RR 1.05 [0.90–1.22]); and emerging adults (aged 20–29 years), except in Brazil, where adults aged 30–49 years had higher rates than emerging adults (RR 1.29 [1.22–1.36]). Middle and late adults (aged 50 years and older) had a higher relative increase in visit rates during the pandemic compared with the pre-pandemic period in several countries, with significant changes ranging from RR 2.84 (95% CI 2.62–3.07) in females aged 50–69 years in Peru to RR 1.08 (95% CI 1.01–1.16) in males aged 50–69 years in the USA.

Interpretation Our findings underscore the central role of primary care in anxiety and depression management and highlight the need for sustained funding, workforce support, and age-sensitive and sex-sensitive interventions to address both ongoing and crisis-related mental health demands.

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Introduction

Anxiety and depression are two of the most common mental health disorders affecting individuals, families, and

communities globally. The 2021 Global Burden of Disease (GBD) study estimated the prevalence of anxiety in individuals aged 20 years and older at 5.5% and depression at

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Research in context

Evidence before this study

We searched the Cochrane Library and PubMed databases for articles up to July 25, 2025, examining the influence of the COVID-19 pandemic on anxiety-related and depression-related visits to primary care. The search terms included MeSH headings such as “anxiety” OR “depression” AND “primary health care” OR “general practice” AND “COVID-19”; and “primary care” OR “family medicine” AND “coronavirus” OR “SARS-CoV-2” AND “visits” OR “consultations” OR “consults” in the title and abstract, without language restrictions. The search identified 148 unique articles, of which, 14 were relevant evidence before this study.

Identified big data studies explored anxiety-related and depression-related visit outcomes to primary care across the globe and reported conflicting trends in visit rates. Authors stated that the different health systems and pandemic-associated changes in mental health help-seeking behaviours were key drivers of the observed trends. They noted that more research was required to inform future health system resource management and promote public awareness in preparation for the next global health crisis. Other studies focused on the adoption of telehealth and its outcomes on visit rates and quality of care.

Previously, the International Consortium of Primary Care Big Data Researchers examined more than 11.4 million mental health-related visits to primary care from Jan 1, 2018, to Dec 31, 2021. Although the onset of the COVID-19 pandemic was significantly associated with increased mental health-related visits to primary care in six of the nine studied countries, all countries experienced increased rates at some point during the pandemic period. Notably, Canada, Peru, Singapore, and Sweden sustained elevated rates throughout the entire pandemic period. The observed

increase in mental health visit rates was largely driven by a surge of anxiety-related and depression-related visits in all countries considered except for China.

Added value of this study

Our research has extended the duration of the COVID-19 study period to include a post-pandemic recovery period, presenting anxiety-related and depression-related visits to primary care from January, 2018 to December, 2023. We applied a life stage, stratified by sex, epidemiology lens to the analyses, allowing for a deeper understanding of the potential country-specific, psychosocial drivers of anxiety and depression in primary care visits during the pandemic and thus identifying a number of potential areas for targeted population health intervention.

Implications of all the available evidence

To future-proof primary care against the next global pandemic, it needs to be purpose-built to absorb additional mental health care burden and still maintain routine health care; this includes workforce and resource protections to buffer against the competing health priorities. Furthermore, each country produced its own unique pattern of visit trends, implicating systemic disadvantage as a key driver of the mental health visit burden in primary care. If governments truly wish to alleviate this burden, they must prioritise research, policy, and interventions that target the psychosocial determinants of mental health vulnerability. Finally, investment in public health campaigns is crucial to enhance health literacy, address changes in health help-seeking behaviours, and to combat harmful misinformation—and primary care is well placed to deliver such campaigns.

5.8%.¹ Anxiety is characterised by excessive worry, fear, or nervousness about events or activities, and depression by low mood and reduced interest in previously enjoyed activities.² They are also often comorbid conditions.³

Although only 25% of WHO Member States report integration of mental health into primary care,⁴ it remains the first place where people with anxiety or depression present to a health practitioner. Furthermore, the needs of people experiencing anxiety and depression are often complex, with primary care commonly playing a key role in their management and coordination across health-care sectors.^{5,6} Consequently, the primary care treatment burden and cost for patients with these conditions has been estimated at up to two times higher than those who do not experience them.⁷

COVID-19 first appeared in China in late 2019, and by March, 2020, had spread worldwide. In response, many countries implemented strict pandemic mitigation and suppression policies.⁸ Although the outcomes of these policies on anxiety and depression prevalence were inconclusive,⁹ there was clear evidence of an increase in

their burden of disease.^{10,11} In response, a global call was made to mobilise high-quality mental health data to identify population-specific areas of need and health-care infrastructure priorities and opportunities.¹²

During the COVID-19 pandemic, mental health services faced unprecedented challenges, such as reduced access to mental health specialist care while inpatient psychiatric units were repurposed as COVID-19 wards.¹³ As a result, primary care had to rapidly adapt to an increased burden of mental health care provision.¹⁴ Therefore, in order to inform a future response to global health crises, investigation of the mental health-related implications of COVID-19 on primary care is essential.

Previously, the International Consortium of Primary Care Big Data Researchers (INTRePID) demonstrated COVID-19-related increases in primary care visits (from 2018 to 2021) for mental health in nine countries.¹⁵ Here, we leverage additional data from the post-pandemic recovery era for primary care anxiety and depression visits from 11 countries. By further examining pandemic stage, psychosocial life stage, and sex data, we aim to provide a

detailed understanding of international trends in anxiety and depression visits to primary care to inform policy planning for the next global health crisis.

Methods

Study design and data sources

We performed a retrospective, repeated cross-sectional study, analysing trends in primary care visits related to anxiety and depression among adults aged 20 years and older, from Jan 1, 2018, to Dec 31, 2023. The study was conducted across 11 countries: Australia, Brazil, Canada, China, Indonesia, Mexico, Norway, Peru, Singapore, Sweden, and the USA. All available visits were included, with no exclusions based on patients' sociodemographic characteristics or comorbidities. Patients were actively involved in the design and conduct of this study through our working group, which includes members of the Patient and Clinician Engagement Program within the North American Primary Care Research Group. This study was approved by the University of Toronto Research Ethics Board (protocol number 46355), from which a waiver of informed consent was granted because only aggregated, de-identified data were used. INTRPID complies with GATHER standards.¹⁶

Health care utilisation data were sourced from a combination of electronic medical records and administrative health databases. Data source details are provided in the appendix (pp 2–5). Each country contributed aggregated monthly or quarterly data (Norway) on primary care visits for anxiety and depression, as well as total primary care visits. Country-specific data limitations included: Norway only providing quarterly data from Jan 1, 2018, to Dec 31, 2022; Peru and Indonesia not having data available for the year 2018. Further search strategy and selection criteria details are provided in the appendix (pp 29–31).

Procedures

Data were disaggregated by sex (male or female) and four age groups based on a psychosocial life-stage analytic framework:¹⁷ emerging adulthood (age 20–29 years), established adulthood (30–49 years), middle adulthood (50–69 years), and late adulthood (70 years and older).¹⁸ A detailed explanation of the psychosocial life stages is available in the appendix (pp 32–33). Sex was recorded based on routine clinical or administrative data entry. Information on gender identity, race, and ethnicity was not available. Both in-person and remote (telephone or video) consultations were included. Data extraction and aggregation were conducted independently by each country's team using a standardised protocol for consistency. Anxiety-related and depression-related visits were identified using country-specific diagnostic codes. Some datasets captured multiple diagnoses per visit, while others reported only the primary reason. Coding practices remained consistent over time, so observed changes are likely to reflect true variations in visit frequency. A harmonisation process aligned diagnostic coding systems across countries (International

Classification of Primary Care, Second Edition [ICPC-2]; International Classification of Diseases, Ninth Revision, Clinical Modification [ICD-9 CM]; International Classification of Diseases, Tenth Revision, Clinical Modification [ICD-10 CM]; Systematized Nomenclature of Medicine, Clinical Terms [SNOMED CT]; and the Ontario Health Insurance Plan diagnostic codes [OHIP]). The specific coding systems used in each of the 11 countries are detailed in the appendix (pp 2–4). To maximise comparability, we mapped all relevant codes to Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition definitions,² including all items defined under anxiety and depressive disorders. Local researchers (KT, CMH, APPN, LC, WCWW, DZ, APS, AFS, MGAR, BG-P, EETG, K-AW, KEE, MSCF, JS-V, ZJL, LHG, J-AM-N, SKWW, RSK, JMW, and GG-V) reviewed the mappings to confirm that the selected codes are used in practice. Codes for anxiety and depression were combined, as these conditions often overlap clinically and coding practices might lead to their inconsistent or interchangeable use. The list of diagnostic codes used is available in the appendix (pp 6–7).

Statistical analysis

We calculated the proportion of anxiety-related and depression-related visits relative to the total number of primary care visits for each country across the study period. Quarterly rates of anxiety-related and depression-related visits for each sex and life stage combination were plotted to illustrate temporal trends. We presented a histogram showing the distribution by sex and life stage of anxiety-related and depression-related visits across countries.

The study period was divided into three distinct phases: pre-pandemic (Jan 1, 2018–Mar 31, 2020), pandemic (April 1, 2020–Dec 31, 2021), and recovery (Jan 1, 2022–Dec 31, 2023). Due to the different timing of the pandemic in China, its periods were defined as follows: pre-pandemic (Jan 1, 2018–Dec 31, 2019), pandemic (Jan 1, 2020–Sept 30, 2021), and recovery (Oct 1, 2021–Dec 31, 2023).

The primary analysis assessed monthly or quarterly counts of anxiety and depression-related visits, modelled as a rate using the natural logarithm of total monthly visits as an offset. The main exposure was time. Age group and sex were included as covariates and potential effect modifiers. Harms were not assessed, as this study used aggregated, de-identified observational data. To reduce the risk of bias, we included both primary and secondary diagnoses (where available), harmonised coding systems across countries, adjusted for secular trends, and accounted for seasonality and autocorrelation in the models.

Quantitative variables, including age and visit counts, were handled by categorising age into four life stages (20–29 years, 30–49 years, 50–69 years, and 70 years and older). The choice of categories was confirmed using sensitivity analyses (negative binomial regression models), testing alternative age groupings to ensure they captured significant and clinically relevant differences in visit rates. No post-hoc analyses were performed.

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See Online for appendix

We applied generalised linear models with a negative binomial distribution to address overdispersion in the count data. Seasonal variation was modelled using sine and cosine terms with a 12-month cycle when seasonality was present. In countries providing quarterly data (Norway) and those with limited visit counts (China, Indonesia, and Mexico), seasonal effects were captured using quarter-specific indicators (Q1–Q4).

The secular trend, which refers to the underlying long-term change in visit rates over time, independent of short-term fluctuations or specific events, was modelled as a single continuous time variable, month 1–72 (or quarter 1–20 for cases in which only quarterly data were available), to capture long-term changes independent of the three phase indicators (pre-pandemic, pandemic, and recovery). Temporal autocorrelation was assessed using autocorrelation function plots of Pearson residuals. Because visit counts in adjacent months are typically correlated, failing to model this serial dependence can underestimate standard errors or bias estimates. An autoregressive lag-1 [AR(1)] term was included in the model when lag-1 autocorrelation was evident; otherwise, the AR(1) term was not applied. Two modelling strategies were employed per country: a main-effects model estimating changes in visit rates by period, sex, and age group; and a series of stratified models by sex and age group to assess subgroup-specific temporal patterns. Stratified models were not conducted for China, Indonesia, and Mexico due to low visit counts leading to unstable estimates. Estimated marginal means and pairwise contrasts were calculated to derive rate ratios (RRs) and 95% CIs for comparisons across time periods. No formal multiple-testing correction was applied. We did not report *p* values or apply a formal multiple-testing correction across the many pre-specified phase contrasts (overall and within age-defined and sex-defined strata). Our interpretation focused on RRs with 95% CIs and on patterns consistent across strata, rather than on statistical significance.

Missing data for Peru and Indonesia in 2018 were addressed by excluding the unavailable period from the analysis. No sampling weights were applied, as the study included all available primary care visits. Statistical model description and results are provided in the appendix (pp 18–28). All analyses were performed using R version 4.3.0.

Role of the funding source

The funder of the study had no role in the study design, data collection, data analysis, data interpretation, and writing of the report.

Results

Between Jan 1, 2018, and Dec 31, 2023, we examined 1 059 101 897 primary care visits from 11 countries. Of these, 49 022 341 (4.6%) visits were attributed to anxiety and depression among adults aged 20 years and older. The highest proportion of visits were found in Canada (18 240 011 [8.1%] of 224 512 695) and the USA (451 067 [11.3%] of 3 998 109); while the lowest proportions were

identified in Indonesia (11 579 [0.2%] of 6 482 240) and Singapore (109 540 [0.82%] of 13 312 087; figure 1; appendix p 8).

Across all countries, females consistently had more visits for anxiety and depression than males in every life stage (figure 2; appendix pp 9–10). Visit volumes were generally lowest among emerging adults (20–29 years) and late adults (70 years and older), with established adults (30–49 years) accounting for most consultations (figure 2; appendix pp 9–10). Exceptions included Singapore, where emerging adults (age 20–29 years) had the highest visit volumes, and the USA, where adults aged 70 years and older had relatively high visit volumes compared with other countries. When considering the proportion of visits within each sex and life stage, emerging adults (20–29 years) generally had a higher share of anxiety-related and depression-related visits than observed in other groups (appendix p 16).

The regression analysis comparing sex and life stage difference showed that, across all countries, females generally had higher visit rates than males, although the difference was not statistically significant in China, Indonesia, and Norway (figure 3; appendix p 11). Emerging adults (20–29 years old) consistently showed the highest visit rates, except in Brazil, where established adults (30–49 years old) had the highest rates (RR 1.29 [95% CI 1.22–1.36]). A gradual decline in visit rates was observed across all countries among individuals in middle adulthood (50–69 years old) and late adulthood (70 years and older; appendix pp 11, 17).

The regression modelling of period effects showed that all countries experienced an upward trend in anxiety-related and depression-related visit rates during the pandemic period compared with the pre-pandemic baseline. The largest relative increase was observed in Peru (RR 2.17 [95% CI 1.96–2.40]), while the smallest significant increase was seen in Norway (RR 1.06 [1.01–1.11]). Increases in Australia, Sweden, and Mexico did not reach statistical significance (figure 4; appendix p 12).

In the recovery period, most countries exhibited either stabilisation or a decline in anxiety-related and depression-related primary care visit rates relative to the pandemic period. The exception was Sweden, where visit rates continued to rise (RR 1.15 [95% CI 1.08–1.22]; appendix p 12). Compared with the pre-pandemic period, most countries experienced sustained or elevated visit rates during the recovery phase. The highest relative increases were observed in Peru (RR 1.91 [95% CI 1.65–2.22]) and China (RR 1.63 [1.20–2.21]). Increases were also evident in the USA (RR 1.20 [1.12–1.29]), Canada (RR 1.19 [1.11–1.27]), and Sweden (RR 1.18 [1.07–1.30]; figure 4; appendix p 12).

The comparative analysis across periods, stratified by sex and life stage subgroups, showed that in the 20–29-year age group (emerging adults), Australia, Canada, and Peru consistently showed increases during the pandemic in both males and females. Singapore showed significant increases only among females, whereas the USA only showed significant increases among males. By contrast, during the recovery phase, most countries experienced significant

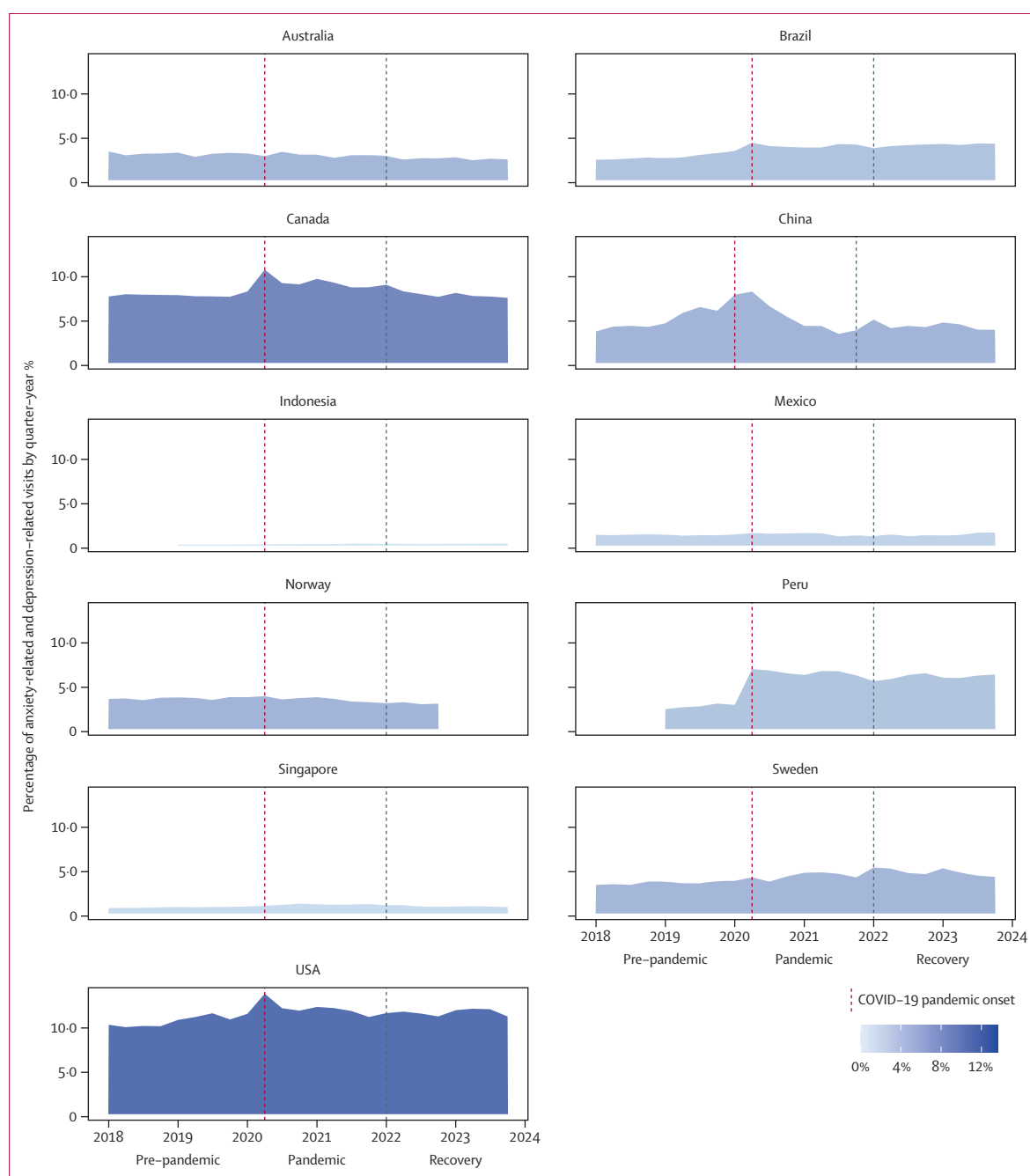


Figure 1: Quarterly rates of anxiety-related and depression-related visits to primary care as a percentage of total visits across the studied periods

The colour gradient reflects the overall proportion of visits attributed to anxiety and depression over the entire study period. Data for 2023 in Norway and for 2018 in Indonesia and Peru were unavailable. Time periods were defined as follows: pre-pandemic, Jan 1, 2018–Mar 31, 2020 (China: Jan 1, 2018–Dec 31, 2019); pandemic, April 1, 2020–Dec 31, 2021 (China: Jan 1, 2020–Sept 30, 2021); recovery, Jan 1, 2022–Dec 31, 2023 (China: Oct 1, 2021–Dec 31, 2023). The red dotted line indicates the onset of the COVID-19 pandemic, and the grey dotted line marks the beginning of the recovery period. These lines delineate the three time periods shown: pre-pandemic, pandemic, and recovery.

declines or stable rates in this age group, compared with the pandemic period (figure 5; appendix pp 13–14).

In the 30–49-year age group (established adults), trends were generally consistent with those observed among younger adults. Canada, Peru, and Singapore showed

substantial increases in both sexes during the pandemic. Brazil, and the USA also showed significant increases in females only. During the recovery, declines were observed in Australia, Brazil, Canada, Peru, and Singapore compared with the pandemic period (figure 5; appendix pp 13–14).

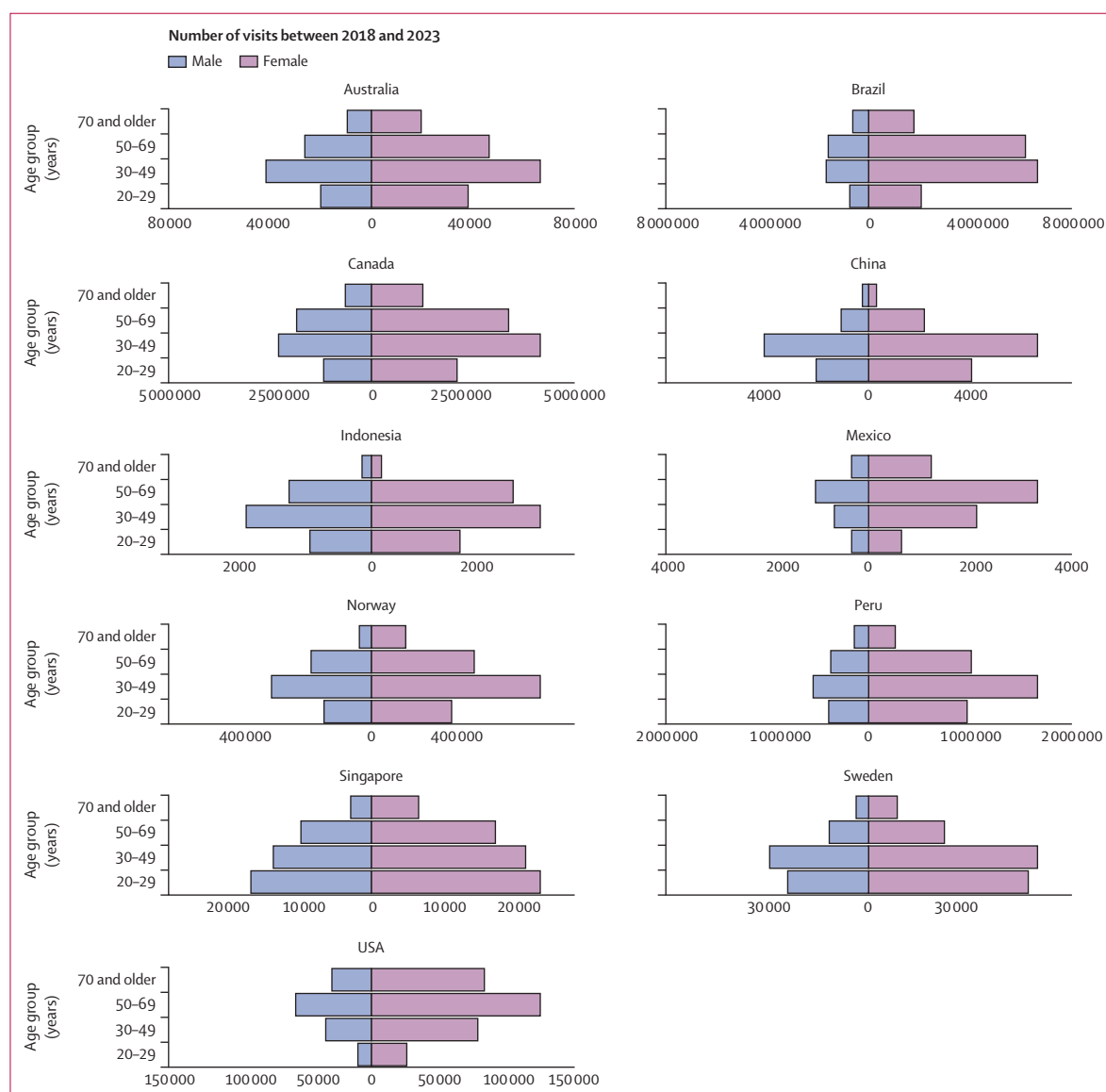


Figure 2: Distribution of anxiety-related and depression-related visits to primary care by sex and life stage across the entire study period
Data for 2023 in Norway and for 2018 in Indonesia and Peru were unavailable. x-axis scales differ across panels.

Among those in the 50–69-year age group (middle adulthood), changes were more pronounced, with large pandemic-related increases in Peru, Brazil, Canada, the USA, and Singapore across both sexes. The largest increases were in Peru (females: RR 2.84 [95% CI 2.62–3.07]; males: RR 2.69 [2.46–2.94]). Norway and Sweden showed increases in females only. In the recovery period compared with the pandemic, significant declines were reported in nearly all settings, except Sweden, which showed increased female visits (RR 1.29 [95% CI 1.02–1.63]; figure 5; appendix pp 13–14).

In the 70 year and older age group (late adulthood), increases during the pandemic period compared with the pre-pandemic period were observed in both sexes in Brazil, Canada, Norway, Peru, and the USA (females ranging from:

RR 1.13 [95% CI 1.01–1.26] to RR 2.10 [1.92–2.29]; and males: RR 1.11 [1.03–1.19] to RR 2.35 [2.12–2.61]); whereas in Singapore, increases were seen in males only. The recovery period was marked by significant reductions in most countries (figure 5; appendix pp 13–14). When comparing the recovery phase to the pre-pandemic baseline, visit rates remained elevated in Brazil (both sexes in those aged 50 years and older), in all age groups in Canada and Peru, in females aged 50+ in the USA, in Singapore in males aged 50–69 years, and in Swedish females aged 50–69 and males aged 70 years and older (figure 5; appendix p 15).

Discussion

Across INTRePID countries, established (30–49 years old) and middle adults (50–69 years old) had the highest

number of primary care visits for depression and anxiety, while females consistently showed higher visit rates than males. Although emerging adults (aged 20–29 years) generally had fewer overall visits, they showed the highest proportion of anxiety-related and depression-related visits compared with other groups. Most countries experienced increases in anxiety and depression visit rates during the pandemic, with many remaining elevated in the recovery period. Through increased awareness of population sex and psychosocial life stage trends in anxiety-related and depression-related visits to primary care before, during, and after the COVID-19 pandemic, we highlight several key psychosocial domains that warrant attention in preparation for future global health crises. Although definitive explanations are beyond our data, these patterns probably reflect a complex interplay between COVID-19, different pandemic responses, and other country-specific influences.

There were a few outlier trends that warrant closer examination. First, in Indonesia, visit trends appeared reasonably consistent but the visit rates were lower than in other included countries. This is consistent with the results of a recent Indonesian Ministry of Health survey where of the 61.1% who received a “not good” mental health screening, only 1.3% made a subsequent mental health-care visit.¹⁹ Beyond barriers such as stigma and lack of awareness, mental health care in Indonesia remains under-resourced, especially in primary care facilities, with limited training²⁰ and low mental health medication availability²¹ creating further barriers to help-seeking.

The visit trend in Peru was also distinct. In early 2020, Peru saw a steep increase in the rate of visits to primary care, which peaked shortly after the onset of the COVID-19 pandemic and remained elevated across the study period. Peru experienced a 106-day national lockdown from March up to the end of June, 2020, and also had the highest confirmed deaths per 100 000 people of all included countries.²² Furthermore, in 2019, the Peruvian government increased access and availability to mental health care via a number of specialist primary care clinics,²³ and in March, 2020, enabled these clinics to provide telemedicine.²⁴ Therefore, the visit trends in Peru might also reflect a successful government-driven integration of mental health care into primary care.

Sweden was another outlier as it showed higher visit rates to primary care during the recovery period than the pandemic period. Sweden also had a far less restricted day-to-day life during the pandemic period compared with other countries. The unique COVID-19 strategy in Sweden relied upon voluntary restrictions rather than imposing population-level strategies. Notably, Swedish trends in antidepressant prescription rates from 2018 to 2023 show a similar trend to their primary care anxiety and depression visit rates, an increase in prescribing rates that was not associated with increased mental health specialist utilisation.²⁵ This suggests that the increase was predominantly contained to Swedish primary care.

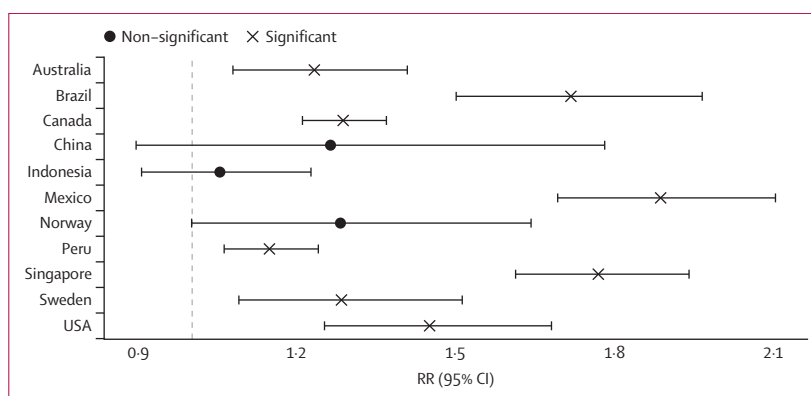


Figure 3: Estimated RRs and 95% CIs for anxiety-related and depression-related visits to primary care in females versus males

The dotted line indicates the null value (RR = 1), representing no difference between females and males. Data for 2023 in Norway and for 2018 in Indonesia and Peru were unavailable. In China, smaller visit counts reduced the precision of the estimates, resulting in wider 95% confidence intervals. RR=rate ratio.

The potential role of the media and misinformation has also been implicated in anxiety and depression visit rates. For example, the negative and sensationalised reporting of the COVID-19 pandemic in the USA media has been associated with poorer mental health outcomes.²⁶ As this was not limited to the USA, there has been a global call to prioritise research on the impact of media and misinformation on mental health during the so-called “COVID-19 infodemic”.¹²

Considering our USA data, after tracking somewhat evenly from 2018 to 2022, the primary care visit rates of emerging adults diverge, with females showing a marked increase in visits compared with males. Notably, the USA Supreme Court overturned *Roe vs Wade* in July of 2022, and its significant effect on the mental health of women aged 20–29 years has since been documented.²⁷ Although further research is required to substantiate such claims, we suggest that these examples demonstrate how the complex dynamics of countries might have potentially affected their anxiety and depression primary care visit trends.

Overall, females aged 20 years and older showed higher visit rate estimates than males. This suggests that while exacerbated by COVID-19, the increased anxiety and depression burden for females exists outside the pandemic. Furthermore, established (30–49 years old) and middle-adult females (50–69 years old) were responsible for the greatest overall volume of visits to primary care.

Previous studies have consistently shown a higher anxiety and depression burden for females of all ages compared with males. These include GBD study data from 1990–2021^{10,18} and 2019–2021.¹¹ The various studies suggested that the increased burden was due to female psychosocial associated factors. For example, the critical transition years of emerging adult females; their increased likelihood of being the main provider of childcare and care of older people, often as a single parent or in lower socioeconomic circumstances;⁹ and their vulnerability to interpersonal abuse and violence.¹¹

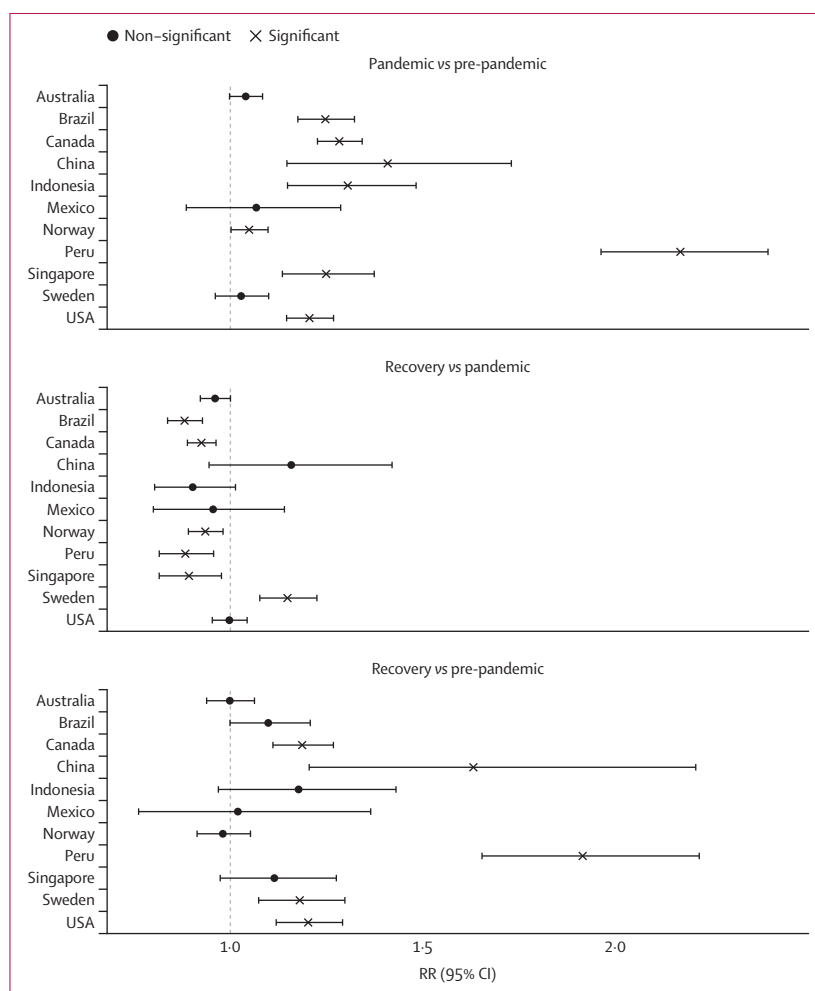


Figure 4: Estimated RRs and 95% CIs for anxiety-related and depression-related visits to primary care across studied periods

The dotted line indicates the null value (RR = 1), representing no difference between periods. Data for 2023 in Norway and for 2018 in Indonesia and Peru were unavailable. Time periods were defined as follows: pre-pandemic, Jan 1, 2018–Mar 31, 2020 (China: Jan 1, 2018–Dec 31, 2019); pandemic, April 1, 2020–Dec 31, 2021 (China: Jan 1, 2020–Sept 30, 2021); recovery, Jan 1, 2022–Dec 31, 2023 (China: Oct 1, 2021–Dec 31, 2023). RR=rate ratio.

Therefore, the heightened burden of anxiety and depression in females has been a global trend for at least three decades and was only exacerbated by the COVID-19 pandemic. Furthermore, not only do females experience a greater anxiety and depression burden across the life course, but it seems even more prominent during a global pandemic. This highlights an urgent need for interventions specifically targeting the psychosocial needs of women if we are to reduce their mental health visit burden on primary care.

When the data were stratified by sex and psychosocial life stage, most countries showed a distinct and unique trend, or so-called footprint, for anxiety and depression visits to primary care. This was most apparent in the visit distributions by life stage and sex from 2018 to 2023 and suggests that the drivers of these trends might be addressed, at least partly, by large-scale interventions. For example, established and middle adults were seen to make

the most visits to primary care when compared with other life stages. This is consistent with anxiety and depression GBD data from 1990 to 2021 for ages 30 to 49 years, and 50 to 69 years.¹⁸ From the psychosocial perspective, established (30–49 years) and middle-adult (50–69 years) life stages are defined by significant financial, career, romantic, and parental competing priorities. Therefore, reducing the psychosocial burden of established and middle adults might also reduce their visit load to primary care.

Although emerging (20–29 years) and established adults (30–49 years old) showed higher consultation volumes, late adults (70 years and older) consistently accounted for a lower number of visits for these conditions. This trend is likely multifactorial. First, older adults often live with multimorbidity and complex health needs, so primary care visits might be driven by other conditions, with anxiety and depression less frequently recorded as the primary reason for consultation.¹⁸ Second, under-recording is possible when physicians prioritise management of physical health concerns over mental health issues. Finally, barriers to accessing care—such as reduced mobility, reliance on others for transport, or limited use of telecommunication technologies—might have further contributed to lower recorded rates in this group, particularly during the pandemic.¹⁴

The regression analysis showed that overall, emerging adults had the highest visit rate for primary care visits compared with other age groups, except in Brazil, where adults aged 30–49 years had the highest rates. The psychosocial features of emerging adults include moving out of home, undertaking tertiary education, entering the workforce, and finding a life partner. These inform potential areas of intervention that might reduce their anxiety and depression burden. Another important consideration is the increased vulnerability of emerging adults (20–29 years) to the effects of social media and misinformation, which might be further elevating the rates of this lifestage.²⁸ This highlights the need for future global health crisis responses to consider misinformation mitigation strategies, for example public health campaigns.¹² Additionally, given the necessity of primary care to continue to provide routine care during a global crisis, primary care is well placed to play an important role in the delivery of public health campaigns aimed at countering misinformation.

Of note, during the pandemic, the relative increase in visit rates was higher among middle adults (50–69 years old) and in some cases late adulthood (70 years and older) in Brazil, Canada, Peru, and the USA, suggesting a potential shift in service utilisation patterns toward older populations amid that phase. Isolation, greater vulnerability to COVID-19, and separation from loved ones are some reported factors associated with these trends.²⁹ Another consideration is that older adults are more likely to experience lower concentrations of health literacy and thus, the conflicting health messaging of the infodemic might be linked to their poorer mental health outcomes.³⁰ These factors again highlight the need for effective public health campaigns during a global pandemic.

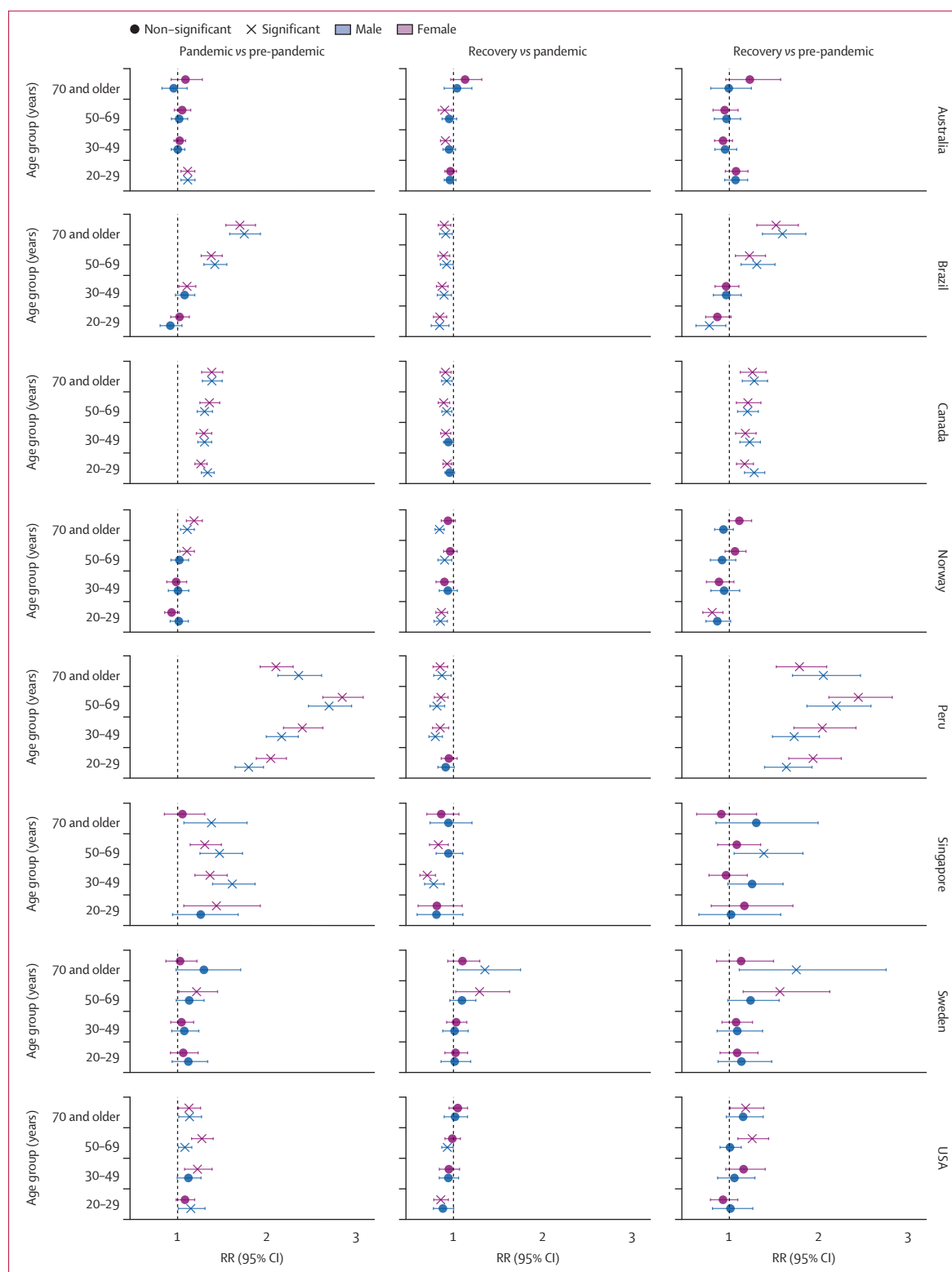


Figure 5: Stratified RRs and 95% CIs of anxiety-related and depression-related primary care visits across periods

The dotted line indicates the null value (RR=1), representing no difference between periods. Data for 2023 in Norway and for 2018 in Peru were unavailable. Time periods were defined as follows: pre-pandemic, Jan 1, 2018–Mar 31, 2020; pandemic, April 1, 2020–Dec 31, 2021; recovery, Jan 1, 2022–Dec 31, 2023. Stratified analyses were not conducted for China, Indonesia, and Mexico due to low counts in subgroups, which resulted in unstable estimates. RR=rate ratio.

A major strength of this study is its use of more than 1 billion primary care visits across 11 countries, encompassing a wide diversity of cultural contexts, health system structures, and attitudes towards mental health. To our knowledge, this is the first multi-country study spanning the Americas, Europe, Asia, and Oceania, to examine trends in anxiety-related and depression-related primary care visits across the pre-pandemic, pandemic, and post-pandemic periods; and the first study to apply a sex and psychosocial life stage analytic framework.

Our study has several limitations. First, the structure and scope of primary care data varied across countries, with some providing national coverage and others only regional samples, limiting generalisability. Findings from countries with regional data (eg, China, Singapore, and Sweden) might not be fully representative of the national population. Results from countries with lower visit counts (China, Indonesia, and Mexico) or shorter observation periods (Norway) should be interpreted with caution due to reduced estimate precision and increased susceptibility to sampling variability. Although each country shows unique patterns, a meta-level comparative analysis could reveal broader trends and should be explored in future research incorporating medical-sociology and qualitative perspectives.

Second, the accuracy of mental health coding depends on documentation quality and could be influenced by cultural perceptions of mental illness, stigma, and clinical coding practices, increasing the likelihood of under-reporting in some countries. Alternatively, in Canada diagnostic coding for anxiety might be applied more broadly, potentially inflating recorded rates, as OHIP includes only one anxiety diagnosis code (compared with more than 80 in ICD-10 CM). Cultural stigma around anxiety is also lower, and physicians can charge more for time-based mental health visits than for regular visits, creating an incentive to use that code. More generally, coding systems varied across countries (ICPC-2, ICD-9 CM, ICD-10 CM, SNOMED CT, OHIP) and in some instances over time, and differences in their specificity and orientation might have introduced heterogeneity that could affect cross-country comparisons.

Third, the data reflect visit counts rather than unique patients, so repeated visits by the same individual cannot be distinguished. In addition, the sex variable refers to biological sex as captured in data sources and might not reflect gender identity. Our analysis involved multiple comparisons across time periods, age groups, and sexes within each country without formal multiplicity adjustment, which might increase the likelihood of chance findings. Results should be interpreted with caution, with inference prioritising effect sizes, 95% CIs, and patterns that are consistent across strata and countries rather than isolated significant results. Cross-country contrasts are descriptive rather than causal.

Fourth, during the COVID-19 pandemic, the workload and availability of primary care providers might have influenced visit volumes and the composition of visits. Factors such as involvement in vaccination campaigns,

increased administrative tasks, staff shortages, and periods of illness or quarantine could have affected service utilisation independently of patient needs.¹⁴

Finally, the categorisation of age groups into life stages might oversimplify the nature of psychosocial development, and while we used a widely recognised framework, the choice of cutoffs might have influenced the age patterns. Our available data were structured in 5-year or 10-year bands; therefore, definitions were adapted to ensure consistency and preserve sufficient sample sizes.

In conclusion, this study showed evidence on how anxiety and depression primary care visits from 11 countries had unique distribution patterns or footprints. The outcomes highlight the value of large-scale interventions for reducing the mental health burden on primary care. The observed trends offer insights into policy planning and guide future research to identify the areas of greatest impact. We identified the most vulnerable groups requiring additional support in the event of a future global pandemic and offer potential pathways to their better mental health and wellbeing.

Collaborator list

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Contributors

AF, MCL, APPN, LC, KM, AH, and KT conceived and designed this study, interpreted the data, and critically reviewed the Article for important intellectual content. MCL performed the data analysis. KT was responsible for funding acquisition and project administration and supervision. AF, MCL, and KT drafted the original work. CMH, APPN, WCWW, DZ, APS, AFS, MGAR, BG-P, EETG, K-AW, KEE, MSCF, JS-V, ZJL, LHG, J-AM-N, SKWW, RSK, JMW, and GG-V acquired and curated the data from their respective countries, interpreted the data, and critically reviewed and edited the manuscript for important intellectual content. MK interpreted the data and critically reviewed the manuscript for important intellectual content. All authors approved the final version of the manuscript and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work were appropriately investigated and resolved. MCL, KT, and JS-V directly accessed and verified the underlying data reported in the manuscript. All authors confirm that they had full access to all the data in the study and accept responsibility for the decision to submit for publication.

Declaration of interests

KT received grants from the Rathlyn Foundation Primary Care Big Data Research and Innovation Fund. KM is an employee of AXDEV Group, with no financial, supervisory, or intellectual involvement in this work. KM serves on the NAPCRG Board and receives reimbursement for travel, lodging, and meals for organisational activities unrelated to this study. KM is member of the PaCE Committee and receives reimbursement for attendance at unrelated meetings, with no payment related to this work. KM serves on Passerelle's Patient Advisory Committee, with no payments or involvement in this study. KM holds an ongoing visiting researcher status at Toronto Metropolitan University, with no involvement in this work. KM is enrolled as a student at the Ontario College of Art & Design University, with no institutional role or contribution to this study. APS received grant funding from the Australian Medical Research Future Fund, the National Health and Medical Research Council Australia, and the National Institute for Health and Care Research (NIHR) National Health Service AI Award. APS has received consulting fees from Tyree Institute of Health Engineering and payment or honoraria from *Annals of Family Medicine*. APS also receives support for attending

meetings or travel from the American Thoracic Society, Lancet *Global Health Commission on People-Centered Care for Universal Health Coverage*, and The George Institute for Global Health. APS has held multiple leadership and fiduciary roles, including serving as Past Chair of the Primary Care Special Interest Group of the Thoracic Society of Australia and New Zealand, and currently serving as Associate Editor of the *Annals of Family Medicine*; Chair of the Membership Committee for the Society for Medical Decision Making; and Commissioner for the Lancet *Global Health Commission on People-Centred Care for Universal Health Coverage*. MSCF receives payment for research work from Universidad Peruana Cayetano Heredia (Peruvian Cayetano Heredia University), as well as grant funding from the NIHR Impact Salud project. MK has received funding support from The Ian Potter Foundation and the University of New South Wales. All researchers are independent of funders. All other authors declare no competing interests.

Data sharing

Data used in this study are subject to data sharing agreements between the University of Toronto and partnering institutions. These agreements permit analysis within the University of Toronto but do not allow public sharing of the data. The analytic code is available upon request from the corresponding author (with publication) and will be shared with interested researchers upon reasonable request.

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